

MITOCHONDRIAL DISORDERS

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OXIDATIVE PHOSPHORYLATION

Final stage of cellular respiration where the cell produces most of its ATP(energy) using electrons from NADH & FADH₂.

Location: inner mitochondrial membrane

Net ATP: 2 ATP per glucose molecule

Inherited Defects in Oxidative Phosphorylation

13

Of the ~90 polypeptides required for oxidative phosphorylation are encoded by mtDNA

~77

Polypeptides encoded by nuclear DNA

mtDNA mutates 10 times faster than nuclear DNA.

Maternal Inheritance

mtDNA is inherited exclusively from maternally

Vulnerable Tissues

Tissues with highest ATP demand (brain, nerves, retina, muscle, heart) suffer severe impairment.

Table 6.1: Disorders of Mitochondrial Oxidative Phosphorylation

Disease	Characteristics
Kearns-Sayre syndrome	• Weakness or paralysis of eye muscles with drooping eyelids (ptosis), vision loss, cardiac conduction defects, unsteadiness when walking (ataxia), muscle weakness in limbs, kidney problems, deterioration of cognitive function (dementia), and short stature • Features appear before age 20 • Caused by mutation in mtDNA
Leber hereditary optic neuropathy (LHON)	• Bilateral central vision loss caused by retinal detachment • Onset usually in the patient's 20s or 30s • Caused by mitochondrial inheritance along maternal line, however four times more males are affected than females

Leigh disease

- Severe neurologic disorder that manifests in first year of life. Progressive swallowing problems, poor weight gain, hypotonia, weakness, ataxia, nystagmus, and optic atrophy accompany lactic acidosis
- Death usually occurs between ages 2 and 3 yrs from respiratory failure
- Caused by mutations in nuclear or mtDNA

Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS)

- Progressive neurodegeneration
- Repeated episodes of lactic acidosis and myopathy
- Cells often contain mutant and wild-type mtDNA and expression is variable

Myoclonic epilepsy
with ragged-red
fibers (MERRF)

- Progressive condition
- Uncontrolled muscle contractions, dementia, ataxia, and myopathy
- Caused by mutation in mtDNA; expression of disease is variable

Neuropathy, ataxia,
and retinitis pigmen-
tosa (NARP)

- Progressive condition
- Sensory neuropathy with numbness or tingling in the extremities, muscle weakness, ataxia and vision loss, cognitive decline, and seizures
- Caused by mutation in mtDNA altering ATP synthase and reducing ability to make ATP

CLINICAL CASE STUDY

Patient History

A 22-year-old man develops painless blurred central vision in one eye. About 8 weeks later, the same problem occurs in the other eye.

Family history: Two maternal uncles became blind when they were young adults.

Molecular Etiology

- Mutation in mitochondrial DNA (mtDNA) affects the ND4 gene of Complex I.
- Complex I cannot transfer electrons from NADH to ubiquinone normally.
- Low ATP damages retinal ganglion cells and causes optic nerve degeneration.

High-Yield Features

- More common in young men (about 80% of affected patients).
- Retina and CNS are vulnerable because they need a lot of ATP.
- Fundoscopy may show optic disc swelling and telangiectatic vessels.

Key point: Mitochondrial DNA is maternally inherited because sperm mitochondria are destroyed after fertilization.

ANSWER:

Leber Hereditary Optic Neuropathy (LHON)



THANK
YOU

